Smart Occupancy
Monitoring System

Automated Bidirectional Counting & Real-Time Cloud Monitoring via LoRaWAN

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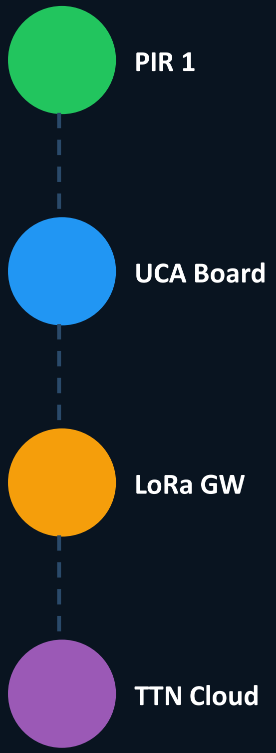


Project Context & Objectives

Smart Building IoT

Automated real-time room occupancy tracking for smart university building management.

Bidirectional Tracking

Entry/Exit detection using two Passive Infrared (PIR) sensors with directional FSM logic.

Low-Power LoRaWAN

Long-range wireless aggregation at 868 MHz with no dependency on local Wi-Fi infrastructure.

Regulatory Compliance

Strict respect of European 1% Duty Cycle radio constraint via enforced TX_INTERVAL guard.

Embedded Hardware Integration

SLIDE 03

1 **UCA Board (ATmega328PB)**

Core microcontroller — 868 MHz RF stage

2 **PIR 1 — Outside Sensor**

VCC → Pin A2 | Signal → Pin A3

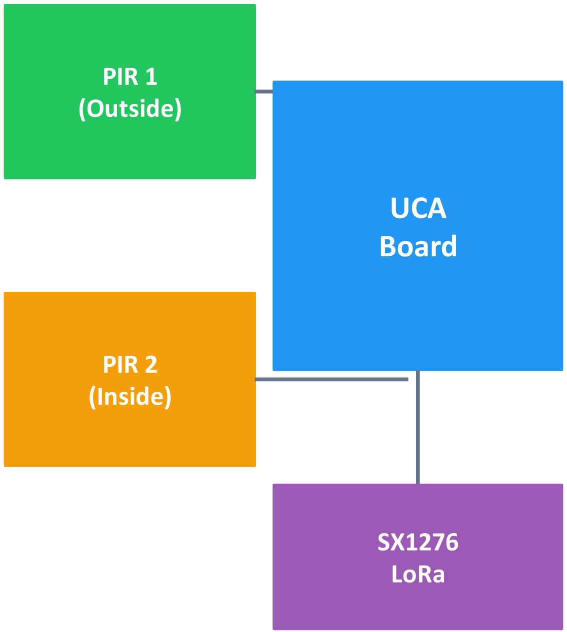
3 **PIR 2 — Inside Sensor**

VCC → Pin A0 | Signal → Pin A1

4 **SX1276 LoRa Transceiver**

Integrated RF module, tuned 868 MHz antenna

System Wiring Overview



Resolving Sensor Crosstalk — The Paper Blinder Solution

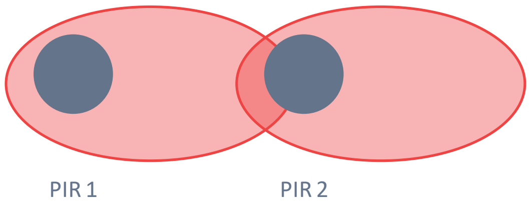
⚠ The Problem

- Standard PIR sensors: 110° wide field of view
- Side-by-side placement → overlapping detection fields
- Simultaneous triggers → directionality blocked
- Counter increment/decrement impossible to determine

✓ Hardware Fix

- Custom cylindrical collimation tubes (blinders)
- Material: rigid black opaque paper, hand-fabricated
- Restricts each sensor to a narrow parallel beam
- Ensures clean sequential cutoff — enables FSM direction logic

Before Fix



After Fix

Non-Blocking Directional Algorithm (FSM)

IDLE

Waiting for first trigger

WAIT_CONFIRM

800ms window open

EVENT_COUNTED

Count +1 or -1

COOLDOWN

2000ms rebound filter

```
// Entry Detection
if (PIR1_HIGH) {
  t1 = millis();
  if (PIR2_HIGH &&
     millis()-t1 < 800) {
    peopleCount++;
  }
}

// Exit Detection
if (PIR2_HIGH) {
  t2 = millis();
  if (PIR1_HIGH &&
     millis()-t2 < 800) {
    peopleCount--;
  }
}
```

Optimized Network Transmission — OTAA & Payload Design

OTAA Activation

- Secure 3-key handshake
- DevEUI — unique device ID
- AppEUI — app identifier
- AppKey — encryption root
- Dynamic session keys per join

TX Guard

10s minimum TX_INTERVAL
enforced by scheduler loop

3-Byte Payload Structure

Byte 0	Byte 1	Byte 2
0x65	0xNN	0x0A

Network Flow

UCA Board

SX1276 RF

LoRa GW

TTN V3

Dashboard

TTN Live Data Validation — Campus Valrose

OTAA	3B	868	TTN V3
Join Mode	Payload Size	MHz Band	Network Server

```
// TTN V3 Console — Live Uplink Log

09:42:11  [LORA] Handshake Complete! DevEUI joined network successfully.
09:42:43  [TX] Uplink sent → Payload: 65 01 0A | RSSI: -82 dBm | SF: 7
09:43:14  [TX] Uplink sent → Payload: 65 02 0A | Entry confirmed. Count: 2
09:43:45  [TX] Uplink sent → Payload: 65 01 0A | Exit confirmed. Count: 1
```

- Stable OTAA join cycles confirmed via serial monitoring
- 3-byte uplinks decoded correctly on TTN V3 developer console
- Bidirectional counting accuracy validated through manual entry/exit tests

Conclusion & Future Enhancements

✓ Delivered

- Functional end-to-end IoT prototype
- Dual PIR + FSM — accurate bidirectional counting
- OTAA LoRaWAN connection to TTN V3
- Compact 3-byte payload design
- Paper blinder fix for PIR crosstalk

→ Next Steps

- I2C OLED display for local visual feedback
- TagoIO Webhook integration for live dashboard
- Multi-entry support for large spaces
- Automated capacity alert thresholds

Merci de votre attention !

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